

BANK ACCOUNT MANAGEMENT SYSTEM USING CLASSES AND OBJECTS

PROGRAM

// UPI Payment and Digital Wallet System

// 1. PaymentService Interface

interface PaymentService

{

void pay(String receiverUPI, double amount) throws
InvalidUPIException,

InvalidAmountException, InsufficientBalanceException;

void checkBalance();

}

// 2. Custom Exception - Invalid UPI

class InvalidUPIException extends Exception

{

public InvalidUPIException(String message)

{

super(message);

}

```
}
```

// 3. Custom Exception - Invalid Amount

```
class InvalidAmountException extends Exception
```

```
{
```

```
    public InvalidAmountException(String message)
```

```
    {
```

```
        super(message);
```

```
    }
```

```
}
```

// 4. Custom Exception - Insufficient Balance

```
class InsufficientBalanceException extends Exception
```

```
{
```

```
    public InsufficientBalanceException(String message)
```

```
    {
```

```
        super(message);
```

```
    }
```

```
}
```

// 5. Wallet Class

```
class Wallet
{
    // Private data members demonstrate encapsulation
    private String userName;
    private String mobileNumber;
    private String upild;
    private double balance;

    // Constructor
    public Wallet(String userName, String mobileNumber,
                  String upild, double balance)
    {
        this.userName = userName;
        this.mobileNumber = mobileNumber;
        this.upild = upild;
        this.balance = balance;
    }

    // Add money to wallet
    public void addMoney(double amount)
    {
        if (amount > 0)
        {
            balance = balance + amount;
        }
    }
}
```

```
        System.out.println("Rs." + amount + " added successfully.");
    }
    else
    {
        System.out.println("Invalid amount!");
    }
}
```

```
// Get balance
public double getBalance()
{
    return balance;
}
```

```
// Get UPI ID
public String getUpild()
{
    return upild;
}
```

```
// Deduct money from wallet
public void deductMoney(double amount)
{
    balance = balance - amount;
}
```

```
}

// Display wallet details
public void displayWalletDetails()
{
    System.out.println("\n----- Wallet Details -----");
    System.out.println("User Name    : " + userName);
    System.out.println("Mobile Number : " + mobileNumber);
    System.out.println("UPI ID      : " + upid);
    System.out.println("Balance     : Rs." + balance);
}
}
```

```
// 6. UPIpayment Class implements PaymentService
class UPIpayment implements PaymentService
{
    private Wallet wallet;

    // Constructor to associate Wallet with UPIpayment
    public UPIpayment(Wallet wallet)
    {
        this.wallet = wallet;
    }
}
```

```
// Pay method

@Override

public void pay(String receiverUPI, double amount)
    throws InvalidUPIException,
        InvalidAmountException,
        InsufficientBalanceException
{
    // Validate UPI ID using String operations
    if (receiverUPI == null ||
        !receiverUPI.contains("@") ||
        receiverUPI.startsWith("@") ||
        receiverUPI.endsWith("@"))
    {
        throw new InvalidUPIException(
            "Invalid UPI ID! Payment cancelled.");
    }

    // Validate amount
    if (amount <= 0)
    {
        throw new InvalidAmountException(
            "Invalid amount! Amount must be greater than zero.");
    }
}
```

```
// Check sufficient balance
if (amount > wallet.getBalance())
{
    throw new InsufficientBalanceException(
        "Insufficient balance! Payment cannot be processed.");
}

// Deduct payment amount
wallet.deductMoney(amount);

System.out.println("\nPayment Successful!");
System.out.println("Receiver UPI : " + receiverUPI);
System.out.println("Amount Paid : Rs." + amount);
}

// Check balance
@Override
public void checkBalance()
{
    System.out.println("Available Balance : Rs."
        + wallet.getBalance());
}
}
```

// 7. Main Class

public class Main

{

public static void main(String[] args)

{

 // Create a wallet

 Wallet userWallet = new Wallet(

 "Gayatri",

 "9876543210",

 "gayatri@upi",

 5000.00

);

 // Create UPIpayment object and associate wallet

 UPIpayment payment = new UPIpayment(userWallet);

 // Add money

 userWallet.addMoney(2000);

 // Display initial wallet details

 userWallet.displayWalletDetails();

```
// Check balance
payment.checkBalance();

// Payment processing using try-catch-finally
try
{
    // Make UPI payment
    payment.pay("friend@upi", 1500);
}
catch (InvalidUPIException e)
{
    System.out.println("Error: " + e.getMessage());
}
catch (InvalidAmountException e)
{
    System.out.println("Error: " + e.getMessage());
}
catch (InsufficientBalanceException e)
{
    System.out.println("Error: " + e.getMessage());
}
finally
{
    System.out.println("\nTransaction process completed.");
}
```

```

    }

    // Display final wallet details
    userWallet.displayWalletDetails();

    // Display final balance
    payment.checkBalance();
}
}

```

OUTPUT

```

PS C:\Users\GAYATRI\eclipse-workspace\Java\src\javacode\java> & 'C:\Program Files\Java\jdk-25.0.3\bin\jav
a.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\GAYATRI\AppData\Roaming\Code\User\workspa
ceStorage\dec763e4a366cbff00288a806970918f\redhat.java\jdt_ws\java_e1c50f8f\bin' 'Bank'
Rs.2000.0 added successfully.

----- Wallet Details -----
User Name      : Gayatri
Mobile Number  : 9876543210
UPI ID        : gayatri@upi
Balance       : Rs.7000.0
Available Balance : Rs.7000.0

Payment Successfull!
Receiver UPI   : friend@upi
Amount Paid    : Rs.1500.0

Transaction process completed.

----- Wallet Details -----
User Name      : Gayatri
Mobile Number  : 9876543210
UPI ID        : gayatri@upi
Balance       : Rs.5500.0
Available Balance : Rs.5500.0
PS C:\Users\GAYATRI\eclipse-workspace\Java\src\javacode\java>

```